Customer Purchase Intelligence

From Transactions to Strategy — analyzing **3,900 retail transactions** to uncover what drives revenue and loyalty.

PYTHON POSTGRESQL POWER BI

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The Business Problem

Who drives revenue?

Identify highest-value demographics

Do discounts work?

Attract valuable customers or bargain hunters?

Who subscribes?

Target the right audience

What to promote?

Products, categories & segments

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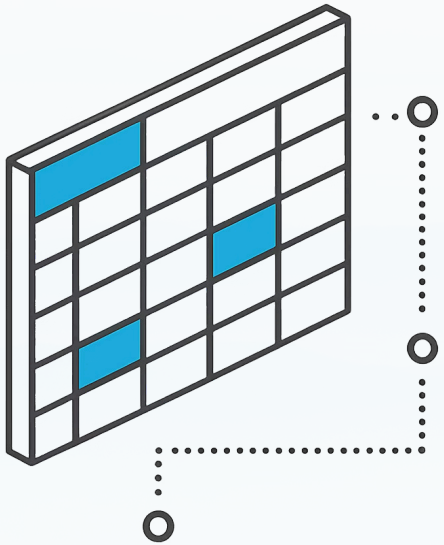
Dataset Overview



3,900 Transactions

18 variables per record

Customer Profile

Age, gender, location, subscription

Purchase Details

Product, category, amount, season

Behavior Data

Discounts, frequency, shipping, ratings

37 missing values found in review ratings — handled during preparation

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Data Preparation (Python)

Quality Check

Impute Missing

Feature Eng

Drop Redundant

Load PostgreSQL

Median imputation preserved category-level trends. Age groups (Teen, Young Adult, Adult, Senior) enabled demographic segmentation.

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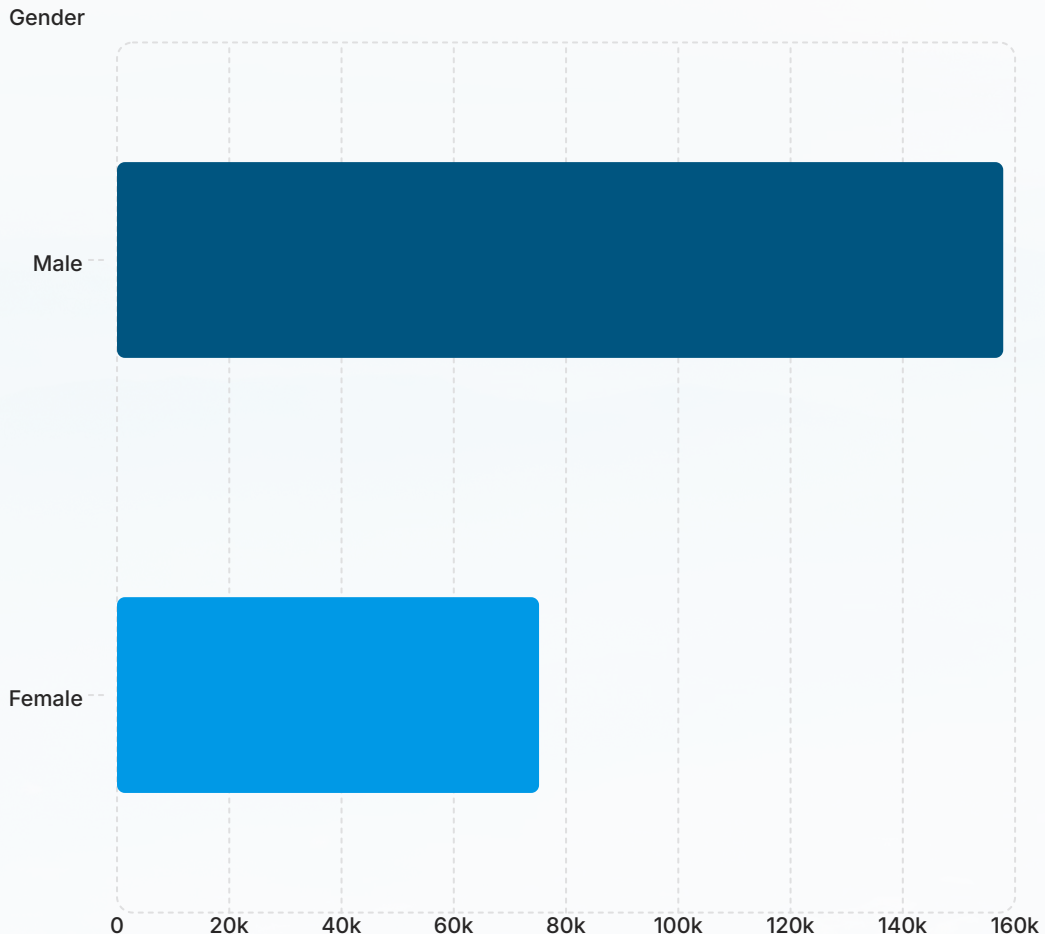
SQL Analysis: Revenue Drivers



Key Findings

- **Male customers** generate 2× more revenue
- **839 discount users** spent above average
- Express shipping: **$60.48** avg vs. Standard **$58.46**

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Products & Discounts

Top-Rated Products

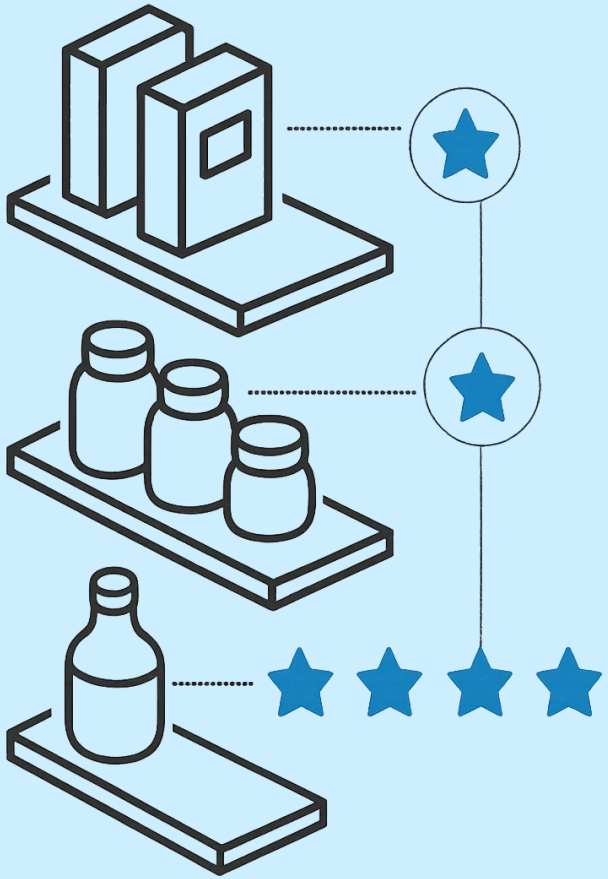
Gloves (3.86★), Sandals (3.84★), Boots (3.82★) — strong promotion candidates

Discount Dependency

Hat 50%, Sneakers 49.66%, Coat 49.07% — margin risk signals

Best-Sellers by Category

Blouse, Jacket, Sandals, Jewelry lead their categories



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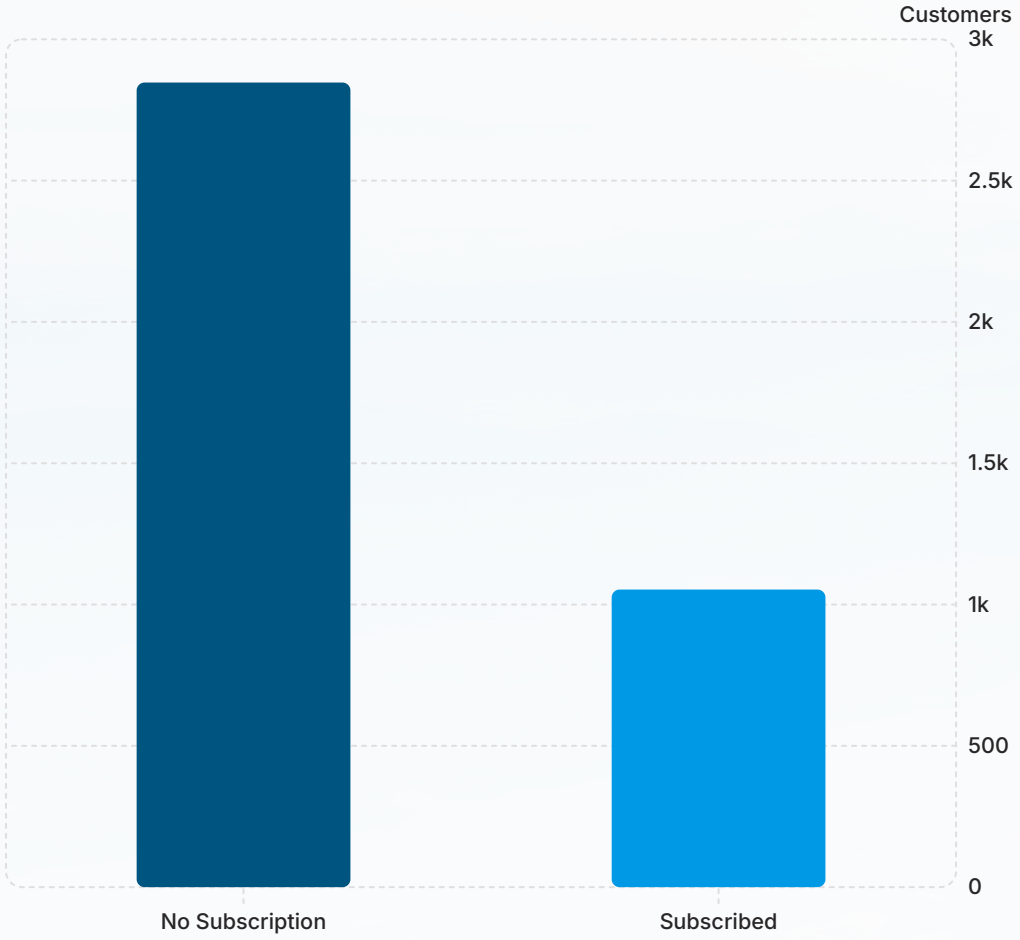
Subscriptions & Customer Segments



Subscription Insights

- Subscribers: **$59.49** avg spend, **$62.6K** total revenue
- Non-subscribers: **$59.87** avg, **$170.4K** total
- **958 repeat buyers** without subscription — conversion opportunity

Customer Segments

- Loyal (>5 purchases): **3,116**
- Returning (2–5): **701**
- New: **83**

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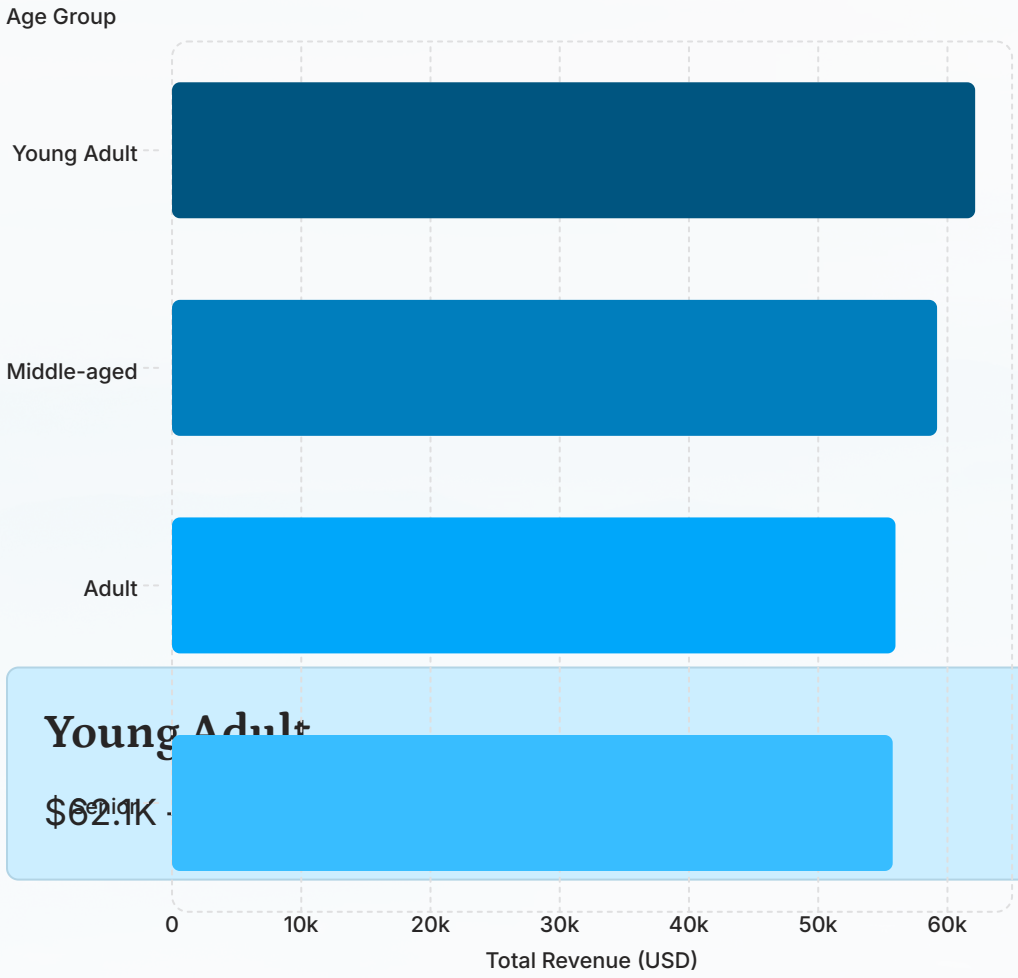
Revenue by Age Group



Targeting Opportunity

Young Adults lead revenue contribution. Disproportionate share suggests focused advertising can maximize ROI.

Young Adult

$62.1K

Senior

$55.8K — growth potential

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Power BI Dashboard

Interactive dashboard for non-technical stakeholders — filters by subscription, gender, category, and shipping.

3.9K Customers

$59.76 Avg Purchase

3.75 Avg Rating

27% Subscribers

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Key Insights & Recommendations

Insights

- Subscribers have higher order value
- Express shipping → more spend
- Discount-heavy products risk margins

Recommendations

- **Grow subscriptions** — target repeat buyers
- **Loyalty program** — reward returning customers
- **Optimize discounts** — reduce blanket promotions
- **Promote top-rated products**
- **Invest in high-value segments**

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